Postdoctoral Fellowship Application

NSF Atmospheric and Geospace Sciences Postdoctoral Research Fellowship

Department of Geography, University of Florida

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Current and Pending (Other) Support

Current Support

University of Florida Graduate Research Assistantship. Mostafa Rezaali is currently supported as a graduate research assistant in the Department of Geography, University of Florida. This support is associated with doctoral research and training in climate science, artificial intelligence for climate extremes, hydroclimatology, and spatiotemporal environmental prediction. The assistantship supports activities such as data preparation, model development, literature synthesis, forecast verification, manuscript preparation, and participation in the intellectual life of the department.

Role: Graduate Research Assistant and Ph.D. candidate.
Sponsoring organization: University of Florida.
Approximate effort: Graduate research appointment during doctoral training.
Relationship to AGS-PRF proposal: The assistantship supports doctoral-stage research and training. It is expected to end before or upon transition to the NSF AGS-PRF fellowship if the fellowship is awarded.

Pending Support

NSF AGS-PRF Proposal: Probabilistic Prediction of Heat Waves and Flash Droughts Using Physics-Informed Conditional Flow Matching on an Icosahedral Mesh. This pending proposal requests support for an independent postdoctoral fellowship project focused on subseasonal prediction of heat waves and flash droughts using a GraphCast-style icosahedral mesh model and conditional flow matching. The project will test whether global atmospheric context, teleconnection indices, land-surface fields, and local persistence can be combined in a probabilistic graph framework to improve calibrated forecasts of U.S. heat extremes at extended lead times.

Role: Proposed Postdoctoral Fellow.

Proposed host context: Department of Geography, University of Florida.

Research focus: Probabilistic, physics-informed GeoAI for atmospheric and climate extremes.

Requested support: Fellowship stipend and fellowship allowance consistent with the AGS-PRF solicitation and Research.gov budget entry.

Project period: Two years, with the requested start date entered in Research.gov.

Other Current or Pending Support

No other current or pending sponsored research support, in-kind support, or overlapping proposal support is known from the materials available for this application draft. This statement should be reviewed and updated before final submission if any additional proposals, appointments, in-kind resources, consulting commitments, foreign or domestic support, laboratory resources, equipment access, travel support, consulting arrangements, or institutional commitments must be disclosed under NSF requirements.

In-Kind Contributions and Facilities

The proposed fellowship relies on normal host-institution research infrastructure, computing access, scholarly mentoring, and professional-development support rather than a separately committed in-kind award. Any formal allocation of computing, data storage, personnel time, travel support, laboratory access, software licenses, or institutional funds beyond standard host support should be added here before final submission if such commitments are made.

Overlap Statement

The pending NSF AGS-PRF proposal is intended to support an independent postdoctoral research program in probabilistic subseasonal heat-extreme prediction. It does not duplicate the graduate assistantship support, which is associated with doctoral training and current research activities at the University of Florida. The fellowship project is forward-looking, postdoctoral in scope, and centered on building a CFM-based forecasting framework and professional-development pathway. The graduate assistantship is a current training and research appointment that should not overlap in effort or budget with the fellowship period if the award is made.

Certification Note

This draft is prepared from the information available locally for application assembly. NSF may require Current and Pending (Other) Support to be generated and certified through SciENcv. Before final submission, the applicant should review this content against the official NSF disclosure requirements and confirm that all support, in-kind resources, and pending proposals are accurately represented.